

7.7 PAM: M-ARY BASEBAND SIGNALING FOR HIGHER DATA RATE

- Binary baseband modulations transmit one bit of information per pulse over the interval of T_b second, or at the bit rate of $R_b = 1/T_b$ bit per second.
- Higher data rate can be achieved by shortening T_b
- For example, to increase the bit rate by M , T_b must be reduced by the same factor of M .
- Increasing the bit rate by M will increase the required channel bandwidth by M .
- A very effective practical solution to increase data rate is to allow each pulse to carry multiple bits.
- By increasing the number of bits per pulse to M bits, the information transmitted by each symbol will also increase by M .

- For example, when $M = 4$ (4-ary, or quaternary), we have four basic symbols, or pulses, available for communication. A sequence of two bits can be transmitted by just one 4-ary symbol.
- This is because a sequence of two bits can form only four possible sequences (viz., **00**, **01**, **10**, and **11**).
- Since with 4-ary, we have four distinct symbols available, we can assign one of the four symbols to each of these combinations (Fig. 7.15a).
- This signaling allows us to transmit each pair of bits by one 4-ary pulse (Fig. 7.15b).
- Hence, to transmit n bits, we need only $(n/2)$ 4-ary pulses.
- Also, because three bits can form $2 \times 2 \times 2 = 2^3 = 8$ combinations, a group of three bits can be transmitted by one 8-ary symbol.

- This special M -ary signaling is known as the ***pulse amplitude modulation (PAM)*** because the data information is conveyed by varying pulse amplitude.
- As M increases, the transmitted power also increases. This is because to have the same noise immunity, the minimum separation between pulse amplitudes should be equal to that of binary pulses. Therefore, pulse amplitudes increase with M .
- It can be shown that the transmitted power increases as M^2 . Thus, to increase the rate of communication by a factor of $\log_2 M$, the power required increases as M^2 .
- Because the transmission bandwidth depends only on the pulse rate and not on pulse amplitudes, the bandwidth is independent of M .

Example 7.4

Determine the PSD of the quaternary (4-ary) baseband signaling in Fig. 7.28 when the message bits **1** and **0** are equally likely.

The 4-ary line code has four distinct symbols corresponding to the four different combinations of two message bits. One such mapping is

$$a_k = \begin{cases} -3 & \text{message bits 00} \\ -1 & \text{message bits 01} \\ +1 & \text{message bits 10} \\ +3 & \text{message bits 11} \end{cases} \quad (7.56)$$

Therefore, all four values of a_k are equally likely, each with a chance of 1 in 4. Recall that

$$R_0 = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_k a_k^2$$

Within the summation, 1/4 of the a_k will be ± 1 , and ± 3 . Thus,

$$R_0 = \lim_{N \rightarrow \infty} \frac{1}{N} \left[\frac{N}{4}(-3)^2 + \frac{N}{4}(-1)^2 + \frac{N}{4}(1)^2 + \frac{N}{4}(3)^2 \right] = 5$$

On the other hand, for $n > 0$, we need to determine

$$R_n = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_k a_k a_{k+n}$$

To find this average value, we build a table with all the possible values of the product $a_k a_{k+n}$:

Possible Values of $a_k a_{k+n}$				
$a_k \backslash a_{k+n}$	-3	-1	+1	+3
-3	9	3	-3	-9
-1	3	1	-1	-3
+1	-3	-1	1	3
+3	-9	-3	3	9

From the foregoing table listing all the possible products of $a_k a_{k+n}$, we see that each product in the summation $a_k a_{k+n}$ can take on any of the following six values $\pm 1, \pm 3, \pm 9$. First, $(\pm 1, \pm 9)$ are equally likely (1 in 8). On the other hand, ± 3 are equally likely (1 in 4). Thus, we can show that

$$R_n = \lim_{N \rightarrow \infty} \frac{1}{N} \left[\frac{N}{8}(-9) + \frac{N}{8}(+9) + \frac{N}{8}(-1) + \frac{N}{8}(+1) + \frac{N}{4}(-3) + \frac{N}{4}(+3) \right] = 0$$

As a result,

$$S_x(f) = \frac{5}{T_s} \implies S_y(f) = \frac{5}{T_s} |P(f)|^2$$

Thus, the M -ary line code generates the same PSD shape as binary polar signaling. The only difference is that it utilizes 5 times the original signal power.

The quaternary (4-Ary) PAM is used by the subscriber loop portion of Integrated service digital network (ISDN). It uses NRZ pulses to transmit 160 kbit/s of data at the **baud** rate (pulse rate) of 80 kbit/s.

7. 8 DIGITAL CARRIER SYSTEMS

- For baseband digital modulation, signals are transmitted directly without any frequency shift in spectrum.
- Because baseband signals have sizable power at low frequencies, they are suitable for transmission over a wireline such as a twisted pair of wires or coaxial cables. Much of the modern communication takes place this way.
- However, baseband signals cannot be transmitted over a radio link or satellites because this would demand impractically large antennas to efficiently radiate the low-frequency spectrum of the baseband signal. Hence, the signal spectrum must be shifted to a high-frequency range.
- A spectrum shift to higher frequencies is also required to transmit multiplex several messages by sharing the large bandwidth of the transmission medium.

- The spectrum of a signal can be shifted to a higher frequency by applying the baseband digital signal to modulate a high-frequency sinusoid (carrier).

7.8.1 Basic Binary Carrier Modulations

- Amplitude Modulation $\longrightarrow$ Amplitude Shift Keying (ASK).
- Frequency Modulation $\longrightarrow$ Frequency Shift Keying (FSK).
- Phase Modulation $\longrightarrow$ Phase Shift Keying (PSK).
- The unmodulated carrier is **$\cos \omega_c t$**
- The baseband signal **$m(t)$** (the modulating signal) can be written according to Eq. (7.1) as

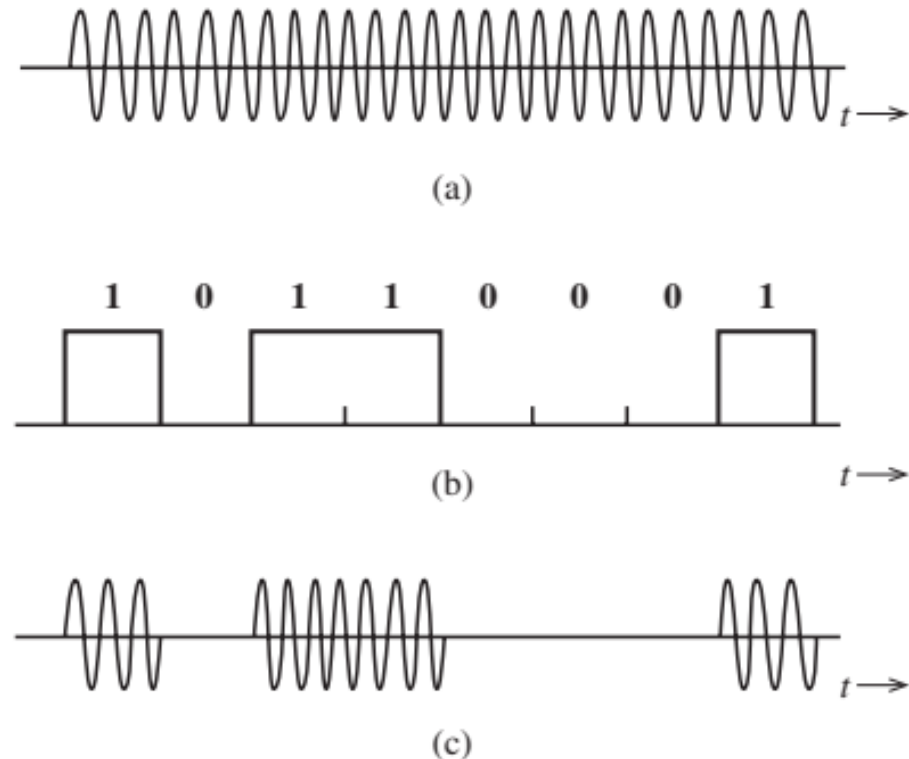
$$m(t) = \sum a_k p(t - kT_b), \quad \text{where} \quad p(t) = \Pi \left(\frac{t - T_b/2}{T_b} \right)$$

- ASK: For ASK, the modulating signal $m(t)$ is on-off with $a_k = 0, 1$, and the modulated signal is

$$\varphi_{\text{ASK}}(t) = m(t) \cos \omega_c t \quad 7.31$$

The ASK is shown in Fig. 17.16. Note that because the modulated signal is an on-off signal, this modulation scheme is also known as on-off keying (OOK).

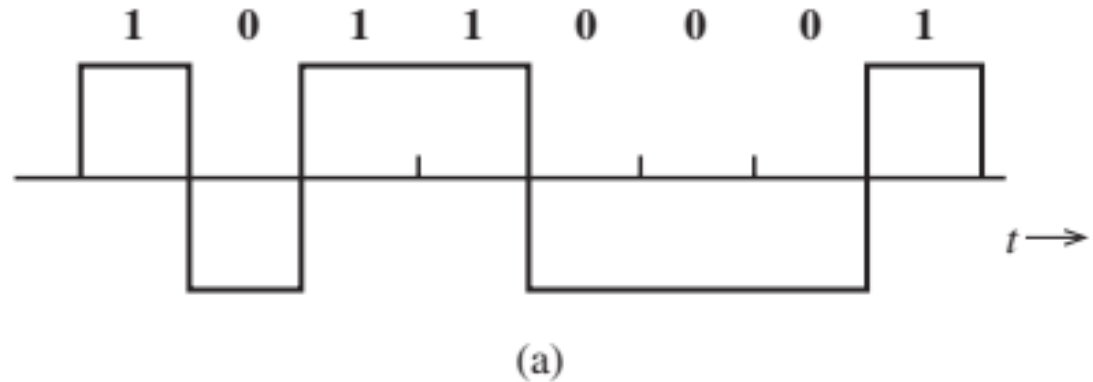
Figure 7.16 ASK:
(a) The carrier $\cos \omega_c t$. (b) The on-off modulating signal $m(t)$. (c) The modulated ASK signal $m(t) \cos \omega_c t$.



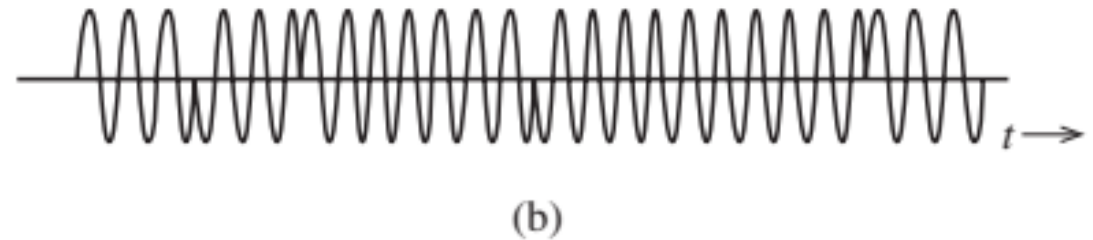
- PSK: If the modulating signal $m(t)$ is polar (Fig. 7.17a), the corresponding modulated signal $m(t)\cos\omega_c t$ would appear as shown in Fig. 7.15b.

Figure 7.17

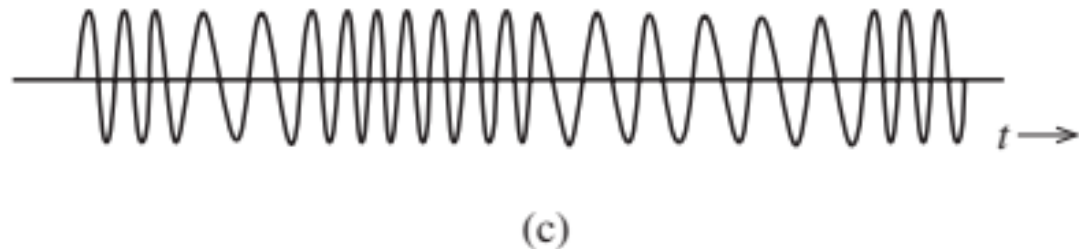
(a) The modulating signal $m(t)$.



(b) The PSK modulated signal



(c) FSK signal modulated signal



- In this case, if $p(t)$ is the basic pulse, a 1 is sent by the pulse $p(t)\cos\omega_c t$ and 0 by $-p(t)\cos\omega_c t = p(t)\cos(\omega_c t + \pi)$. Hence, the two pulses are π radians apart in phase.
 - This means that the information resides in the phase or the sign of the pulse. So, this scheme is known as **PSK**.
 - The PSK modulated carrier signal has the same form as the ASK signal [Eq. 7.31] with the difference that the line code for PSK is polar $a_k = \pm 1$, whereas the line code for ASK is unipolar $a_k = 0, 1$.
- **FSK:** The FSK signal may be viewed as a sum of two interleaved ASK signals, one with a modulating frequency ω_{c0} , and the other with a modulating frequency ω_{c1} as shown in Fig.7.17c. Thus the FSK modulated signal can be represented as

$$\varphi_{\text{FSK}}(t) = \sum a_k p(t - kT_b) \cos \omega_{c1} t + \sum (1 - a_k) p(t - kT_b) \cos \omega_{c0} t \quad 7.32 \quad 12$$

where $a_k = 0, 1$ is on-off line code.

- Thus the FSK signal is a superposition of two AM signals with different carrier frequencies and different but complementary amplitudes.

➤ Notes

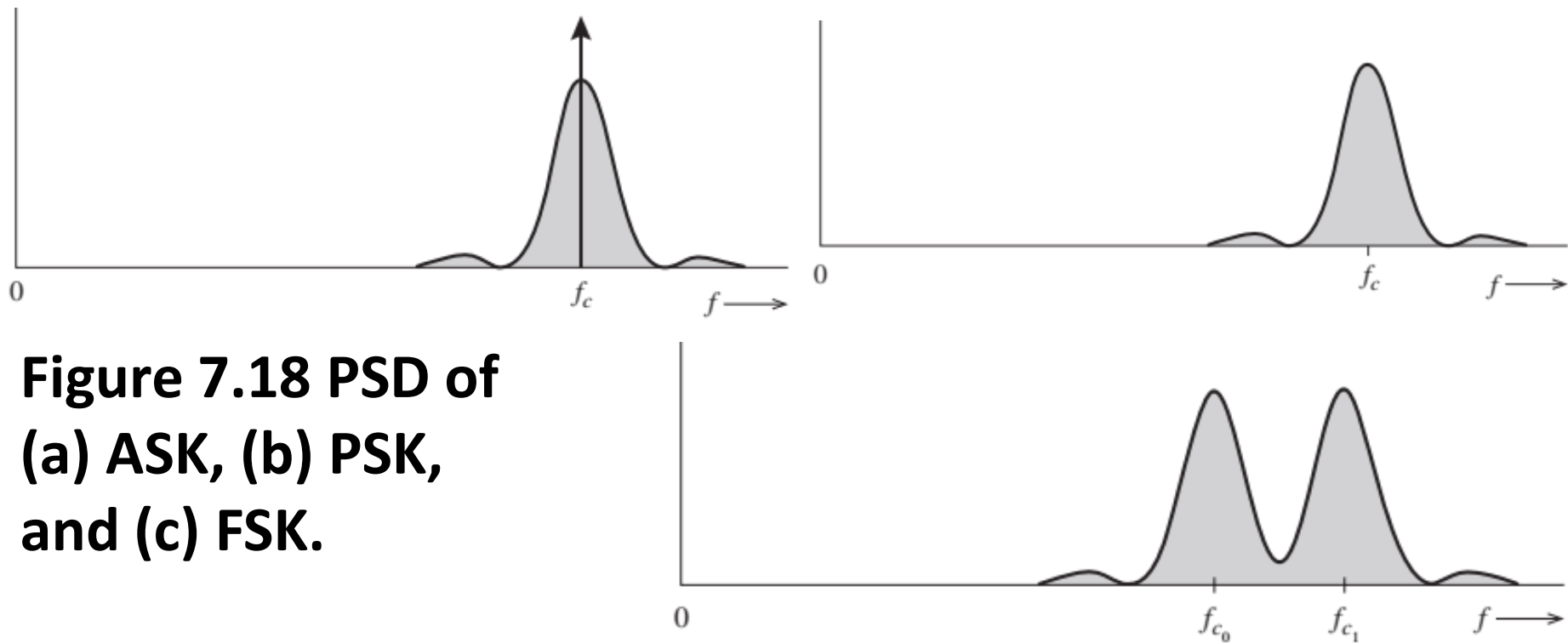
- In practice, ASK as an on-off scheme is commonly used today in optical fiber communications in the form of laser-intensity modulation.
- PSK is commonly applied in digital satellite communications and was also used in earlier telephone modems (2400 and 4800 bit/s).
- For FSK, AT&T in 1962 developed one of the earliest telephone-line modems called 103A; it uses FSK to transmit 300 bit/s at two frequencies, 1070 and 1270 Hz, and receives FSK at 2025 and 2225 Hz.

7.8.2 PSD of Digital Carrier Modulation

- Since the modulation shifts the spectrum of the modulating signal, the PSD of the ASK signal is the same as that of an on-off signal (Fig. 7.8) shifted to $\pm f_c$ as shown in Fig. 7.18a.
- Remember that by using a full-width rectangular pulse $p(t)$,

$$P\left(\frac{n}{T_b}\right) = 0 \quad n = \pm 1, \pm 2, \dots$$

- In this case, the baseband on-off PSD has no discrete components except at dc in Fig. 7.8. Therefore, the ASK spectrum has discrete component only at f_c .



**Figure 7.18 PSD of
(a) ASK, (b) PSK,
and (c) FSK.**

- The PSK signal also fits this modulation description where $m(t)$ is a polar signal using a full-width NRZ pulse. Therefore, the PSD of a **PSK** signal is the same as that of the polar baseband signal shifted to $\pm f_c$, as shown in Fig. 7.18b. Note that this PSD has the same shape (with a different scaling factor) as the PSD of the ASK minus its discrete components.

- Finally, as the FSK signal is a sum of two interleaved ASK signals using the full-width pulse. Hence, the spectrum of FSK is the sum of two ASK spectra at frequencies ω_{c0} and ω_{c1} , as shown in Fig. 7.18c. It can be shown that by properly choosing ω_{c0} and ω_{c1} and by maintaining phase continuity during frequency switching, discrete components can be eliminated at ω_{c0} and ω_{c1} . Thus, no discrete components appear in this spectrum.
- It is important to note that the bandwidth of FSK is higher than that of ASK or PSK
- As observed earlier, polar signaling is the most power-efficient scheme. The **PSK**, being polar, requires 3 dB less power than ASK (or FSK) for the same noise immunity, that is, for the same error probability in pulse detection.